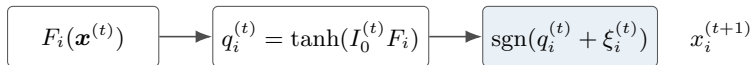
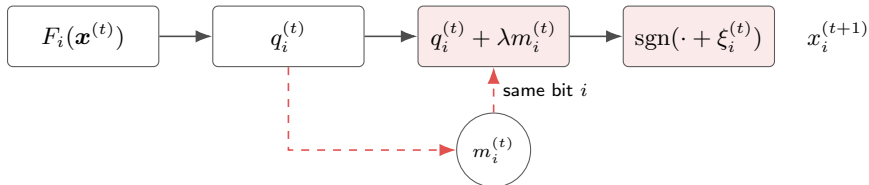


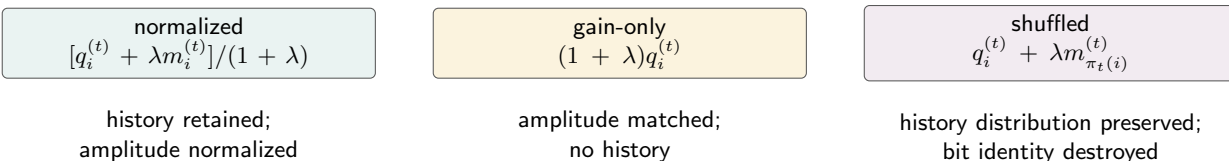
(a) Memoryless pSA



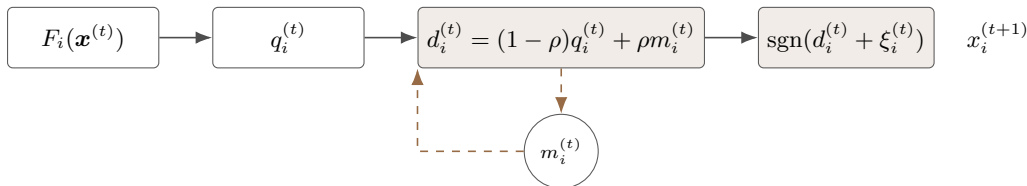
(b) Bit-aligned additive response memory



(c) Controls that isolate temporal reinforcement



(d) Finite-response filtering



$\rho = \exp(-\Delta t / \tau)$ only for an ungated ($p_{\text{hold}} = 0$) ideal single-pole response; $\lambda = 0$ and $\rho = 0$ recover pSA.

The common Bernoulli activation mask commits both x_i and m_i ; it is omitted from each branch for clarity.